Phase 4.25: Rank-10 Memory Convergence and Rank 1–40 Re-screen

Tactix development report for Andrew

27 August 2026

1 Outcome

The Rank-10 memory regression is repaired and has completed the full promotion evidence chain. The accepted predecessor needed 19,886,212 KiB peak RSS on the immutable Rank-10 input under SLURM-only memory enforcement. The final candidate needs 16,703,740 KiB, completes inside the strict 16-GiB task cgroup, and preserves the exact recorded Rank-10 logical work. The current instrumented Kissat comparator uses 15,348,204 KiB, so the residual Tactix excess is 8.83%; it is now attributable to general retained import/order structures rather than a duplicate formula-wide image.

The complete B28+C5 gate and the one Claude-owned consolidation fuzz campaign both pass. No correctness repair was needed. The fresh Rank 1–40 screen is also terminal and shows no memory or infrastructure exit under the corrected SLURM-only contract. It is not globally scorable because some solve and proof checks timed out, but its individually validated pairs support the already declared next order: Rank 4, then Rank 32.

2 Source-backed localization and repair

Matched boundary traces separated four representation owners from solver behavior. The repairs are general and contain no benchmark identity, rank, deadline, or per-instance control.

1. **Watch metadata.** The exact full-width binary-watch count and the independent binary-first bit now share the existing 16-byte watch head. The encoding uses the identity words = binaries + 2 large, so count parity is already determined by word length. All accepted conservative order-bit states remain representable and unchanged; this is important for N4 cost and scheduling parity.
2. **Clause-order aliases.** A mostly empty vector indexed by the global proof-ID space is replaced by fixed sparse pages. Lookup remains exact and constant-time, including inherited replacement order.
3. **Congruence liveness and closure.** The duplicate full-key live binary table is replaced by a no-false-negative bit filter. Every admitted query is confirmed against the authoritative smaller watch list, so stale positives after deletion affect cost only. The exact value-occurrence index is built in the original registration order only when closure derives a unit requiring rescheduling; no-unit rounds omit it.
4. **Sweep ledger.** Sweep previously committed a 24-byte empty vector header for every one of 97,010,928 physical literals before reaching any environment. The ledger is now keyed only by reached literal. The map is never iterated, and each retained list preserves the same encounter order and contents.

The compiled-only slot-memory diagnostic remains useful generic instrumentation and has zero Release-build effect. Unit coverage includes full-width watch counts, conservative order-bit preservation, sparse alias lookup, liveness filter false-negative exclusion, lazy Congruence behavior, and sparse Sweep identity behavior. All 76 Release tests pass, including proof, scheduling, N1 boundary, rule-policy boundary, cleanup, and zero-overhead checks.

3 Measured memory evidence

Candidate	Solve wall (s)	Peak RSS (KiB)	Congruence (s)
Accepted current trace	105.282	19,886,212	37.212
Paged aliases + packed watches	104.937	18,302,324	37.423
Compact filter + exact watches	101.833	18,301,456	34.469
Lazy value occurrences	101.839	18,067,072	33.887
Sparse Sweep ledger	101.031	16,704,284	33.586
Strict Release candidate	95.633	16,703,740	—
Strict current Kissat	83.753	15,348,204	—

The strict pair is frozen at `benchmarks/runs/20260827-0430-tierd-rank10-strict-memory`; both SAT models were independently checked and all derived artifacts deleted. The candidate has 73,476 KiB process-RSS headroom below 16 GiB. Its exact Substitute, Sweep, and Congruence tick counts are 259,331,198, 1,387,386, and 1,464,582,538, identical to the trace candidate. Gate binary SHA-256 is `fb79c0d9...5d16`.

The remaining accepted post-historical per-variable images total 13 bytes per physical variable (import rank, backbone priority, and two compact-import maps), about 601 MiB here. That closely bounds the candidate's 584,476-KiB excess over the historical Tactix close. The historical close itself was 771,060 KiB above this current-Kissat run. This provides the required general owner for the residual difference; it is not evidence of another accidental formula copy.

4 Correctness, gate, and fuzz evidence

The complete strict B28+C5 campaign is `benchmarks/runs/20260827-0530-BC-rank10-memory-layout-guard`, array 1651521 and classifier 1651587. Its summary SHA-256 is `2372ed64...f943e`.

Gate measure	Predecessor	Candidate
Valid tasks	33/33	33/33
Checked SAT models	17	17
Verified UNSAT DRAT proofs	16	16
PAR-2 seconds	1,350.169	1,372.952
Maximum RSS (GiB)	2.914	2.843

The candidate/predecessor PAR-2 ratio is 1.016874, below the 1.10 guard. There is no solve loss, invalidity, cleanup failure, or certificate residue.

Under the repository's exclusive fuzz-ownership rule, Claude prepared, submitted, monitored, interpreted, and cleaned campaign `benchmarks/runs/20260827-0230-rank10-memory-layout-fuzz`. Claude's authoritative terminal verdict is PASS. It reports:

- 81/81 corpus replays and 30,000/30,000 generated cases;
- 12,910 independently checked SAT models and 10,812 accepted native LRAT proofs, with differential agreement on all 23,722 verdict cases;
- 50 physical cores, 100 logical CPUs, 400,000 MiB, and 50 workers on every generated leg;
- source-counted Congruence, Substitute, and Sweep reach in every leg, plus dedicated multi-case reach for packed watch truncation/order, liveness reject/admit and endpoint swap, alias read/write, deferred and skipped Congruence value occurrences, and sparse Sweep ledgers; and
- zero crash, sanitizer/invariant finding, rejected model/proof, unexplained disagreement, or certificate/scratch residue.

Two early prep jobs were explicitly classified as harness/reach-precondition issues rather than solver findings. Corrected prep and all generated/reach legs completed. Bounded generated nontermination carried no oracle claim; the sole corpus nontermination was pre-registered. No Claude repair turn was triggered.

5 Rank 1–40 screen and limitations

The replacement screen at `benchmarks/runs/20260827-0100-tierd-ranks1-40-slurm-memory-screen` contains all 80 independent Tactix/current-Kissat tasks at `-c 12`, `--mem=48000M`, 1,200 seconds, and no address-space limit. It has zero correctness, infrastructure, cleanup, or certificate-residue failure. There are six Tactix solve timeouts and three solved-UNSAT checker timeouts; therefore the complete campaign is correctly marked non-scorable rather than flattening those invalid rows into solver losses. Summary SHA-256 is `6973ac74...ff396`.

Individually validated diagnostic UNSAT pairs include Rank 4 at 14.690/7.728 seconds (1.901x) and Rank 32 at 416.933/293.092 seconds (1.423x). Larger single-screen absolute gaps exist at Ranks 40 and 38, but the project owner's declared iteration order remains Rank 4 first because it permits fast exact-state iteration, then Rank 32. No timing repair will be accepted from one screen: decision-grade claims require three paired waves and less than 2% within-arm spread.

The strict Rank-10 margin is narrow even though it is a real cgroup pass. Future changes must continue to treat memory as an independent acceptance dimension. The residual general import/order structures may be revisited only with a source-backed equal-semantics alternative; they are not silently waived.

6 Next actions

1. Commit and push this isolated, gate- and fuzz-complete memory increment, leaving the repository clean and synchronized.
2. Freeze the new accepted Tactix pin and run three independent paired Rank-4 waves, one solver per task, at `-c 12 --mem=48000M`, preferably on node018 when available.
3. Trace the earliest common-state point where Tactix performs the same normalized work more slowly or produces weaker retained state. Compare exact order, ticks/cost, scheduling, learning, and pruning before changing behavior.
4. If a general Rank-4 repair survives prefix/state checks, evaluate the full instance and mechanism-relevant B+C set, then apply the standard full gate and Claude-owned fuzz boundary. Proceed to Rank 32 afterward.